The Leading One-Loop Hydrogen Lamb Shift: Assembly, a Convention Erratum, and Coulomb-Sturmian Bethe Logarithms

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2026

Abstract

We assemble the hydrogen $2S_{1/2}$–$2P_{1/2}$ Lamb shift at leading one-loop order, $\alpha(Z\alpha)^4$, from the standard expressions reviewed by Eides, Grotch and Shelyuto, with the electron mass in place of the reduced mass and the Bethe logarithms tabulated by Drake and Swainson. The result is 1052.19 MHz, 5.65 MHz (0.534%) below the Lundeen–Pipkin measurement, 1057.845(9) MHz, and 5.64 MHz below the 2019 measurement of Bezginov *et al.*, 1057.8298(32) MHz. This is textbook arithmetic, and the residual is not a measure of missing multi-loop physics: the self-energy part of the one-loop $\alpha(Z\alpha)^5$ term alone, with coefficient $A_{50} = 4\pi(139/128 - \frac{1}{2}\ln 2) \approx 9.29$, adds about +6.9 MHz, more than the residual. Adding the review's other tabulated terms down to the kHz level gives 1057.824 MHz, 0.020 MHz below Lundeen–Pipkin and 0.005 MHz below Bezginov *et al.* The paper records three things. First, an erratum: an earlier assembly used the self-energy constant 38/45 in place of 10/9. The difference is exactly the Uehling coefficient 4/15, which had been subtracted inside the self-energy while the Uehling shift was also added separately; restoring 10/9 moves the Lamb shift by +27.13 MHz. Second, Bethe logarithms computed from scratch in a Coulomb-Sturmian pseudostate basis at exponent Z/n: $\ln k_0(2P)$ and $\ln k_0(3D)$ reproduce the tabulated values to 2.4% and 0.07% with 40 basis functions ($2P$: 1.24% with 55), while the $2S$ acceleration-form value is not converged (-0.92% at 40 functions is a crossing; +2.98% at 50). Third, a negative result: a model two-loop self-energy sum on the Dirac spectrum of the three-sphere grows without bound with its cutoff and yields no finite two-loop coefficient without renormalization counterterms, which it does not contain. Evaluating hydrogen Bethe logarithms through the dynamical symmetry of the Coulomb problem is prior art (Lieber, 1968; Huff, 1969).

1 Introduction

The $2S_{1/2}$–$2P_{1/2}$ splitting of hydrogen, first measured by Lamb and Retherford in 1947 and explained at leading order by Bethe's non-relativistic calculation the same year [1], is the canonical test of bound-state QED. The two direct measurements used here are Lundeen and Pipkin's, 1057.845(9) MHz [2], and the frequency-offset separated-oscillatory-field measurement of Bezginov *et al.*, 1057.8298(32) MHz, obtained from a measured $2S_{1/2}(F{=}0)$–$2P_{1/2}(F{=}1)$ interval of 909.8717(32) MHz after removing the hyperfine contribution [3]. Both are intervals between hyperfine centroids. The modern theory of the interval, order by order, is reviewed by Eides, Grotch and Shelyuto [4]; current work on the two-loop self-energy at low Z is that of Yerokhin, Harman and Keitel [9].

Scope of the claim. The 1052.19 MHz headline is the leading one-loop assembly with tabulated Bethe logarithms. It is not a GeoVac computation and does not compete with precision theory. It omits the one-loop $\alpha(Z\alpha)^5$ term, the reduced-mass factors $(m_r/m)^3$ of the leading terms, recoil, the finite nuclear size, and all two-loop terms, which are added from the review's tables in Sec. 4.1. Nothing in the number uses the three-sphere: no Fock projection, no Dirac spectrum on S^3, no GeoVac label set. What the paper adds is (i) the erratum of Sec. 3, (ii) the Bethe logarithms of Sec. 5, computed from scratch and checked against the Drake–Swainson table [6], and (iii) the two-loop negative of Sec. 6.

Prior art. The use of the hydrogen atom's dynamical symmetry to evaluate Bethe logarithms is due to Lieber, who used the O(4) symmetry together with Green's-function methods to obtain a rapidly convergent series and values for $1S$, $2S$ and $2P$ [7], and to Huff, who recast the calculation in the algebra of SO(4,2) [8]. The Coulomb-Sturmian pseudostate sums of Sec. 5 are a finite basis version of the same spectral sums; we make no novelty claim for the method.

Backing. The assembly and the erratum are recomputed in exact and closed-form arithmetic by the test file `tests/test_paper36_lamb_chain.py`, which also recomputes (slow-marked) the Bethe logarithms from the Sturmian code in `tests/paper36_lamb_support/`. The $\alpha(Z\alpha)^5$ estimate is checked by `tests/test_paper36_a50_term.py`, and the budget of Table 1, with its closure on the review's Table XX, by `tests/test_paper36_lamb_budget.py`. The two-loop model is `geovac/two_loop_self_energy.py`.

2 The one-loop expressions

At order $\alpha(Z\alpha)^4$ and with $m_r = m$, the self-energy and Uehling vacuum-polarization shifts of the $n = 2$ levels are, in Hartree ($\alpha(Z\alpha)^4 mc^2 = \alpha^3 Z^4$ Ha),

$$\Delta E_{\rm SE}(nS_{1/2}) = \frac{\alpha^3 Z^4}{\pi n^3}\Big[\tfrac{4}{3}\ln\frac{1}{(Z\alpha)^2} - \tfrac{4}{3}\ln k_0(n,0) + \tfrac{10}{9}\Big], \tag{1}$$

$$\Delta E_{\rm SE}(2P_{1/2}) = \frac{\alpha^3 Z^4}{8\pi}\Big[-\tfrac{4}{3}\ln k_0(2,1) - \tfrac{1}{6}\Big], \tag{2}$$

$$\Delta E_{\rm VP}(nS) = -\frac{4}{15}\,\frac{\alpha^3 Z^4}{\pi n^3}, \tag{3}$$

with $\Delta E_{\rm VP}(2P) = 0$ at this order. These are the entries of Ref. [4], Sec. VIII.A, Eq. (40) and Table II, rewritten with the common factor $\alpha(Z\alpha)^4 m/(\pi n^3)$: the constant $10/9 = 11/18 + 1/2$ is the Dirac form-factor term ($4\times 11/72$) plus the Pauli form-factor term ($4\times 1/8$); the $2P_{1/2}$ constant $-1/6$ is the Pauli term $4(j(j+1) - l(l+1) - 3/4)/(8l(l+1)(2l+1))$ at $l = 1$, $j = 1/2$; and $-4/15 = 4\times(-1/15)$ is the Uehling term. The Lamb shift is $L = [\Delta E_{\rm SE} + \Delta E_{\rm VP}](2S) - \Delta E_{\rm SE}(2P_{1/2})$.

3 The erratum: a double-counted Uehling term

An earlier implementation of Eq. (1) used the constant $38/45$ in place of $10/9$, and also added the Uehling shift (3). The difference of the two constants is

$$\frac{10}{9} - \frac{38}{45} \;=\; \frac{50-38}{45} \;=\; \frac{4}{15}, \tag{4}$$

exactly the Uehling coefficient. The earlier constant had the Uehling term already subtracted inside the self-energy bracket, so with the separate Uehling shift the vacuum polarization was counted twice. Restoring 10/9 moves the Lamb shift by

$$\Delta L = +\frac{4}{15}\cdot\frac{\alpha^3 Z^4}{\pi n^3}\,\mathrm{Ha} \;=\; +27.13\ \mathrm{MHz\ at}\ n=2,\ Z=1. \tag{5}$$

Observation 1 (Uehling double counting)**.** *With the self-energy constant* 38/45 *and a separate Uehling term the leading one-loop Lamb shift is* 1025.06 *MHz; with the canonical* 10/9 *it is* 1052.19 *MHz. The shift is exactly* $(4/15)\,\alpha^3 Z^4/(\pi n^3)$ *Ha* $= +27.13$ *MHz.*

4 Result

With the Drake–Swainson values $\ln k_0(2S) = 2.8117698931$ and $\ln k_0(2P) = -0.0300167089$ [6], $\alpha^{-1} = 137.035999084$ and 1 Ha = 6 579 683 920.502 MHz (CODATA 2018 [5]):

Quantity	earlier (MHz)	corrected (MHz)	Shift (MHz)
SE $2S_{1/2}$ constant	+38/45	+10/9	+4/15
SE $2S_{1/2}$ total	+1039.31	+1066.44	+27.13
VP $2S_{1/2}$ (Uehling)	−27.13	−27.13	0
Total $2S_{1/2}$	+1012.18	+1039.31	+27.13
SE $2P_{1/2}$	−12.88	−12.88	0
VP $2P_{1/2}$	0	0	0
Lamb shift	+1025.06	+1052.19	+27.13
Lundeen–Pipkin [2]	+1057.845(9)		
Residual	−32.78 (−3.10%)	−5.65 (−0.534%)	

Against Bezginov *et al.* [3] (1057.8298(32) MHz) the corrected residual is −5.64 MHz.

4.1 The residual

The assembly is the leading $\alpha(Z\alpha)^4$ order only, at $m_r = m$. The terms it omits are all tabulated by Eides *et al.* [4], and Table 1 adds them to the $2S_{1/2}$–$2P_{1/2}$ interval. The largest is the one-loop $\alpha(Z\alpha)^5$ term,

$$\Delta E(nS) = \frac{\alpha(Z\alpha)^5}{\pi n^3}\,mc^2\,A_{50}, \qquad A_{50} = 4\pi\Big(\frac{139}{128} - \frac{1}{2}\ln 2\Big) \approx 9.29, \tag{6}$$

about +6.9 MHz on $2S$, or +7.13 MHz with the polarization insertion (Table III of the review). The $2S$ entries are the review's; the $2P_{1/2}$ entries are evaluated from its formulas (one-loop and anomalous-moment terms of Table II, A_{60} and the logarithmic $\alpha(Z\alpha)^6$ terms of Tables IV and V, recoil of Table VIII, and the Barker–Glover $(Z\alpha)^4(m/M)^2$ term).

The budget accounts for the residual. The assembled theory lies 0.020 MHz below Lundeen–Pipkin (2.3σ) and 0.005 MHz below Bezginov *et al.* (1.7σ); the review's own estimate of the uncalculated terms is ±0.002 MHz, and it predates later work on the two-loop terms [9]. None of the added terms is a GeoVac computation.

Term	MHz	Source in [4]
One-loop $\alpha(Z\alpha)^4$ at $m_r = m$ (Sec. 4)	+1052.193	Table II
Reduced-mass factors $(m_r/m)^3$ of the same terms	−1.634	Table II
One-loop $\alpha(Z\alpha)^5$	+7.129	Table III
One-loop $\alpha(Z\alpha)^6$	−0.416	Tables IV, V
One-loop $\alpha(Z\alpha)^7$	+0.002	Tables VI, VII
Two- and three-loop $\alpha^n(Z\alpha)^4$, muonic and hadronic VP	+0.101	Table II
Two-loop $\alpha^2(Z\alpha)^5$ and $\alpha^2(Z\alpha)^6$	−0.041	Tables III, V
Recoil, $(Z\alpha)^5 m/M$ and $(Z\alpha)^4(m/M)^2$	+0.355	Table VIII; Barker–Glover formula
Radiative recoil	−0.002	Table IX
Finite nuclear size, $r_p = 0.8409$ fm	+0.138	Table X
Theory	+1057.824	
Lundeen–Pipkin [2]	+1057.845(9)	
Bezginov *et al.* [3]	+1057.8298(32)	

Table 1: Order-by-order budget of the $2S_{1/2}$–$2P_{1/2}$ interval. At the review's proton radius, 0.862 fm, the same sum is 1057.8315 MHz against its Table XX value 1057.833(2)(4) MHz. Rows are rounded separately; the theory line is the unrounded sum.

5 Bethe logarithms in a Coulomb-Sturmian basis

5.1 Definition and normalization

For a hydrogenic state $|nlm\rangle$ (atomic units),

$$\ln k_0(n,l) = \frac{\sum_k |\langle nlm|\vec{p}|k\rangle|^2 (E_k - E_n) \ln\big(|E_k - E_n|/\mathrm{Ry}\big)}{2Z^4/n^3}, \tag{7}$$

where the sum runs over all intermediate states, bound and continuum, and their magnetic substates. The denominator is the same for every l. For s-states it equals the sum rule $I_v(nS) = \sum_k |\langle nS|\vec{p}|k\rangle|^2 (E_k - E_n) = \frac{1}{2}\langle \nabla^2 V\rangle = 2Z^4/n^3$; for $l > 0$ the corresponding sum vanishes, because $\psi_{nl}(0) = 0$, while the numerator of Eq. (7) stays finite. This uniform normalization is the one in which Eq. (2) and the Drake–Swainson table are written.

In the code the numerator is summed over the initial magnetic substates as well, which multiplies it by $2l + 1$; the code therefore divides by $D(n,l) = 2(2l+1)Z^4/n^3$. The factor $2l + 1$ is this bookkeeping and nothing more: without it the computed $2P$ logarithm would be three times the tabulated one.

5.2 Basis

The radial basis is $\chi_{k,l}(r) = r^{l+1} e^{-\lambda r} L_k^{(2l+1)}(2\lambda r)$, $k = 0, \ldots, N-1$, at $\lambda = Z/n$ for target level n, which contains the exact n-th hydrogenic radial function. All overlap, kinetic, potential, dipole and $1/r^2$ matrix elements are exact polynomial-times-exponential integrals evaluated at 50 digits; the Hamiltonian is diagonalized in the basis after canonical orthogonalization, and the pseudostates of the $l \pm 1$ channels discretize the intermediate spectrum.

5.3 Velocity form for $l > 0$

With $|\langle nl|\vec{p}|k\rangle|^2 = (E_k - E_n)^2\, l_> \,|\langle nl|r|k\rangle|^2$ (summed over substates) the numerator of Eq. (7) converges for $l > 0$ and is evaluated directly. Against the Drake–Swainson values [6]:

- $\ln k_0(2P) = -0.03074$ at $N = 40$, $+2.40\%$ from -0.0300167; -0.03039 at $N = 55$, $+1.24\%$.
- $\ln k_0(3D) = -0.005236$ at $N = 40$, $+0.07\%$ from -0.0052321.

5.4 Acceleration form for s-states

For s-states the velocity form converges slowly. The acceleration form replaces $\vec{p}$ by $\nabla V = Z\hat{r}/r^2$,

$$\ln k_0 = \frac{\sum_k |\langle nl|\nabla V|k\rangle|^2 \ln\big(|E_k - E_n|/\mathrm{Ry}\big)/(E_k - E_n)}{\sum_k |\langle nl|\nabla V|k\rangle|^2/(E_k - E_n)}. \tag{8}$$

In this basis it does not converge:

- $\ln k_0(2S) = 2.7858$ at $N = 40$ (-0.92% from 2.8117699) and 2.8955 at $N = 50$ ($+2.98\%$); the $N = 40$ agreement is a crossing.
- $\ln k_0(1S) = 3.0019$ at $N = 20$ ($+0.60\%$ from 2.9841286) and 3.268 at $N = 40$ ($+9.5\%$); again a crossing.

A converged from-scratch $2S$ logarithm is therefore not available here, and the assembly of Sec. 4 uses the tabulated value.

6 A two-loop negative

A model of the two-loop self-energy was built on the Dirac spectrum of the unit three-sphere [10], $|\lambda_n| = n + 3/2$ with degeneracy $g_n = 2(n+1)(n+2)$, and transverse photon modes of eigenvalue $\mu_q = q(q+2)$ and degeneracy $d_q = q(q+2)$. For the rainbow and crossed topologies it sums

$$\Sigma = \sum_{n_1,n_2,n_3,\,q_a,q_b} \frac{\big(\prod_{i=1}^{4} W_i\big)\, g_{n_1} g_{n_2} g_{n_3}\, d_{q_a} d_{q_b}}{\lambda_{n_1}^2 \lambda_{n_2}^2 \lambda_{n_3}^2\, \mu_{q_a} \mu_{q_b}} \tag{9}$$

over electron levels $0, \ldots, n_{\max}$ and photon levels allowed at each vertex, where $W_i \in \{0, 2\}$ is an SO(4) channel count with the parity rule $n_a + n_b + q$ odd, and the external level is $n_{\mathrm{ext}} = 1$. No bound-state wavefunction enters; the external level stands in for hydrogen $2S$. The dimensionless bracket is $C = \Sigma_{\mathrm{total}}/(4\pi)^2$.

The sum grows without bound: between $n_{\max} = 5$ and 6 the local log-log slope of the total is 2.8 (it was 3.3 under the earlier count, whose totals were 44.8, 266.3, 761.3, 1657.8 and 3052.8). A finite two-loop coefficient requires the wave-function and mass renormalization counterterms, Z_2 and δm, which the model does not contain. The negative is about this model: it does not show that no spectral construction on S^3 can produce the two-loop coefficient, and it makes no comparison with the literature value, since the model has no finite value to compare.

7 Conclusion

The leading one-loop assembly of the hydrogen Lamb shift with tabulated Bethe logarithms gives 1052.19 MHz, 0.534% below the Lundeen–Pipkin measurement. The number is standard one-loop QED with literature inputs. The terms it omits, all tabulated by Eides *et al.*, bring it

Table 2: The model two-loop sum versus cutoff, with the channel count as corrected on 2026-11-07 (an earlier version of the count paired the spinor content of opposite signs of the Dirac operator, and every entry of the table printed before that date was computed with it).

$n_{\max}$	rainbow	crossed	total	C
2	587.4	467.0	1054.4	6.68
3	1680.9	1313.8	2994.8	18.96
4	3653.0	2786.3	6439.3	40.78
5	6761.9	5140.6	11902.5	75.37
6	11265.7	8507.5	19773.2	125.22

to 1057.824 MHz, within 0.020 MHz of Lundeen–Pipkin; the self-energy part of the one-loop $\alpha(Z\alpha)^5$ term alone is about +6.9 MHz of that (+7.13 MHz with the polarization insertion). An earlier version of the assembly double-counted the Uehling term, and the correction is exactly $(4/15)\,\alpha^3 Z^4/(\pi n^3)$ Ha = +27.13 MHz. Bethe logarithms computed from scratch in a Coulomb-Sturmian pseudostate basis reproduce the tabulated $2P$ and $3D$ values to 2.4% and 0.07% at 40 functions, while the $2S$ acceleration-form value has not converged. A model two-loop self-energy sum on the three-sphere Dirac spectrum diverges with its cutoff and gives no finite coefficient without renormalization counterterms. The muonic-hydrogen transfer that an earlier version of this paper reported is in Paper 34.

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